The critical dimension of the one-phase Bernoulli problem

An expository account of the flatness of minimizing cones in dimensions at most six

Abstract. A nonflat one-homogeneous global minimizer of the one-phase Bernoulli energy exists in $\mathbb{R}^7$ (De Silva and Jerison), and none exists in dimensions at most four (Caffarelli, Jerison and Kenig; Jerison and Savin). A recent preprint closes the remaining gap: there are no nonflat minimizing cones in dimensions five and six, so the critical dimension is seven. This note explains how that proof works and why it is organized as it is. After the standard reduction to a cone with a smooth link, the argument is an instability proof: one exhibits a test function for which the second variation of the energy is negative. The test function and an auxiliary vector field are built from the normalized Hessian of the cone, and the pointwise inequalities they must satisfy are inequalities between finitely many tensor variables. These are settled by an explicit algebraic certificate, verified in exact arithmetic. We motivate each step: what the classical Jacobi-subsolution strategy asks for, how "calibrations in jet space" turn it into a finite-dimensional positivity question, why plain powers of the Hessian norm work for cones in $\mathbb{R}^3$ but fail from $\mathbb{R}^4$ on, how a richer ansatz repairs this, and how the singular set of the test function is removed. The same method, with very small certificates, also reproves the classification in dimensions three and four.

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1 Introduction

1.1 The problem

For a ball $B \subset \mathbb{R}^n$ and a nonnegative $v \in H^1(B)$, the *one-phase Bernoulli energy* is

$$J(v; B) = \int_B \left(|\nabla v|^2 + \mathbb{1}_{\{v>0\}} \right) \mathrm{d}x. \tag{1.1}$$

It balances the Dirichlet energy of v against the volume of its positive set. A nonnegative $u \in H^1_{\mathrm{loc}}(\mathbb{R}^n)$ is a *global minimizer* if $J(u;B) \le J(v;B)$ for every ball B and every nonnegative $v \in H^1(B)$ with $v - u \in H^1_0(B)$. Allowing sign-changing competitors changes nothing, because replacing v by its positive part v_+ keeps the boundary values and does not increase the energy. Minimizers are harmonic in the open set $\{u > 0\}$, and on the *free boundary* $\partial\{u > 0\}$ they satisfy, in a weak sense, the Bernoulli condition $|\nabla u| = 1$.

A function is *one-homogeneous* if $u(rx) = ru(x)$ for $r > 0$. The basic examples are the *flat* solutions $u(x) = (x \cdot e)_+$ with $|e| = 1$, and one-homogeneous minimizers are called *minimizing cones*. Let n_* denote the least dimension in which a nonzero, nonflat, one-homogeneous global minimizer exists.

Theorem 1.1 (Critical dimension [1])**.** *Every nonzero one-homogeneous global minimizer of (1.1) in $\mathbb{R}^n$, $1 \le n \le 6$, is flat. There is a nonflat one-homogeneous global minimizer in $\mathbb{R}^7$. Consequently $n_* = 7$.*

The second sentence is the theorem of De Silva and Jerison [2]. The content of Theorem 1.1 is the first sentence in dimensions five and six; dimensions at most four were known [3–5], and we shall see that the same method reproves them.

Minimizing cones govern the singularities of general minimizers. For a local minimizer u in an open set $\mathcal{O} \subset \mathbb{R}^n$ (the same comparison on balls compactly contained in $\mathcal{O}$), a free boundary point x_0 is *regular* if some blowup $\lim u(x_0 + rx)/r$ along $r \to 0$ is a flat solution, and *singular* otherwise; $\mathrm{Sing}(u)$ is the set of singular points.

Corollary 1.2 (Singular set)**.** *Let u be a nonnegative local minimizer of (1.1) in an open set $\mathcal{O} \subset \mathbb{R}^n$. Its free boundary is smooth near every regular point. The singular set is empty if $n \le 6$, locally finite if $n = 7$, and $\dim_{\mathcal{H}} \mathrm{Sing}(u) \le n - 7$ for $n \ge 7$. For every $n \ge 7$ this bound is attained.*

1.2 Context

The variational theory begins with Alt and Caffarelli [3], who proved existence, Lipschitz regularity and nondegeneracy of minimizers, and the regularity of flat free boundaries. Caffarelli's Harnack-inequality approach [6] and De Silva's improvement of flatness [7] give $C^{1,\alpha}$ regularity near regular points; higher regularity goes back to Kinderlehrer and Nirenberg [8], and in the generality needed here is due to De Silva, Ferrari and Salsa [9, 10]. Weiss's monotonicity formula [11] shows that blowups are minimizing cones, and his dimension reduction converts a classification of cones into the bound of Corollary 1.2. A modern account of all of this is Velichkov's book [12].

Thus everything hinges on n_*. It is classical that $n_* \ge 3$ (Alt and Caffarelli). Caffarelli, Jerison and Kenig [4] proved that there are no nonflat minimizing cones in $\mathbb{R}^3$, and Jerison and Savin [5] proved that stable homogeneous solutions with smooth cross-section are flat in dimensions at most four, giving $n_* \ge 5$. In the other direction De Silva and Jerison [2] constructed a nonflat minimizing cone in $\mathbb{R}^7$, so $n_* \le 7$. Further symmetric minimizing cones, including a second example in $\mathbb{R}^7$, were constructed by Firester, Tsiamis and Wang [13], who also showed [14] that their cohomogeneity-one cones with $O(n-k) \times O(k)$ symmetry are unstable for $n \le 6$ and strictly stable for $n \ge 7$. The result discussed here assumes

no symmetry. It concerns cones; rigidity of arbitrary entire stable solutions is a different question, settled in dimension four by Fernández-Real and Serra [15]. For generic boundary data, Fernández-Real and Yu [16] obtained regularity through dimension six by a different route.

The analogy with minimal surfaces is close. There the critical dimension is eight: Simons [17] proved that stable minimal cones with smooth link are flat in dimensions at most seven, by testing the stability inequality with a power of the norm of the second fundamental form, and the Simons cone in $\mathbb{R}^8$ is minimizing. Jerison and Savin developed the Bernoulli analogue, with the Hessian of the cone in the role of the second fundamental form, and explained why powers of the Hessian norm do not give a uniform instability criterion in dimensions four through six [5]; their four-dimensional argument uses a more refined function of the Hessian eigenvalues. The proof discussed here replaces the single test function by a test function together with a vector field, both built from the jet of the cone, and verifies the resulting pointwise inequalities by a finite algebraic certificate.

1.3 The shape of the proof

The argument has four layers, summarized in Figure 1.

1. **Reduction** (Section 2). If all minimizing cones in $\mathbb{R}^{n-1}$ are flat, a nonflat minimizing cone in $\mathbb{R}^n$ is smooth away from the origin, and some component of its positive set has a nonvanishing Hessian. Its link is a smooth domain on the sphere carrying a solution of an overdetermined eigenvalue problem.
2. **What must be produced** (Section 3). Minimality implies a stability inequality on the link. To contradict it one needs a nonnegative "Jacobi subsolution", or more flexibly a test function together with a vector field whose divergence controls the energy of the test function and whose boundary flux has the right sign.
3. **How to find it** (Section 4 to Section 8). Since the cone is unknown, the test function and the field must be universal formulas in the local jet of the cone; the required inequalities then become pointwise inequalities on the space of jets. The simplest choice, a power of the Hessian norm, works only for cones in $\mathbb{R}^3$. A richer ansatz, dictated by the way the simple one fails, leads to polynomial inequalities in finitely many tensor variables, which are proved by an exact algebraic certificate.
4. **Integration** (Section 9). The fields are singular where the Hessian vanishes. A weighted Bochner estimate shows that this set is invisible to the energy, and integration yields a contradiction. Dimension five is reduced to dimension six by a suspension (Section 11).

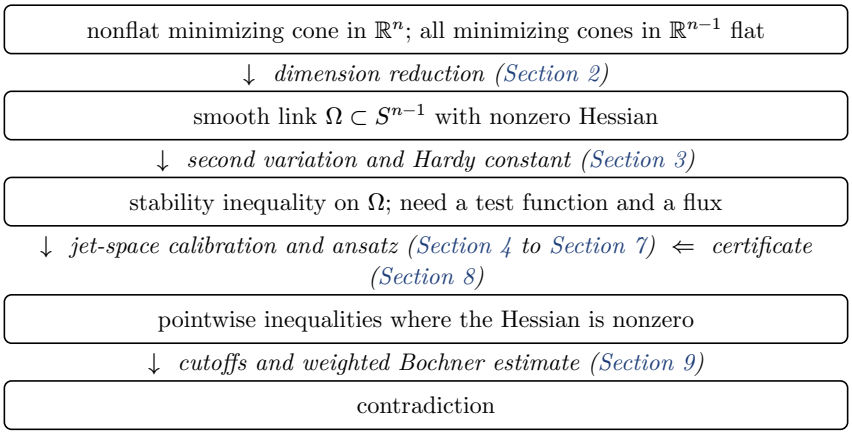


Figure 1: The logical structure of the exclusion of nonflat minimizing cones.

Throughout, the proof is organized as an induction on the dimension: flatness of cones in $\mathbb{R}^{n-1}$ is the input that makes cones in $\mathbb{R}^n$ smooth away from the vertex. Dimension one is elementary, and dimensions two through six are obtained one at a time. The flatness statement of Theorem 1.1 for $n \le 6$ has been formally verified in the Lean proof assistant, including the algebraic certificates, which are checked in exact arithmetic; the verification takes the Lipschitz regularity of minimizers and the smoothness of free boundaries near flat blowups as cited inputs.

1.4 Conventions

On a domain $\Omega \subset S^m$ we use the round metric, covariant derivatives ∇, the Laplace–Beltrami operator Δ, and orthonormal frames in which repeated indices are summed. N denotes the unit normal to $\partial\Omega$ pointing *into* Ω (into the positive phase). For a symmetric tensor, *trace-free* means that every contraction of two slots vanishes, and $\mathrm{STF}^r(\mathbb{R}^m)$ is the space of symmetric trace-free r-tensors. Mean curvatures are sums of principal curvatures, not divided by the dimension. The letter n always denotes the ambient dimension of a cone and $m = n - 1$ the dimension of its link.

2 From minimizers to cones with smooth links

The instability argument at the heart of the proof needs a smooth object to work on: a domain on the sphere with smooth boundary, carrying a function that solves an explicit boundary value problem. A general minimizing cone has no such structure a priori; its free boundary could be singular away from the vertex. The purpose of this section is to show that, if all minimizing cones in one dimension lower are flat, then a nonflat minimizing cone does have this structure. This is the only place where the regularity theory of minimizers enters, and it is what makes the proof an induction on the dimension.

2.1 Regularity inputs

Two questions must be answered about a minimizing cone: what do its blowups at points of the free boundary look like, and when is the free boundary smooth? Regularity theory answers both. We use the following standard facts for a nonnegative local minimizer u in an open subset of $\mathbb{R}^n$, $n \ge 2$ [3, 12].

1. u has a locally Lipschitz representative, harmonic in $\{u > 0\}$.
2. At a free boundary point x_0, every sequence $r_j \to 0$ has a subsequence along which $u(x_0 + r_j x)/r_j$ converges locally uniformly; every such limit is a nonzero one-homogeneous global minimizer (nondegeneracy, compactness, and Weiss monotonicity [11]).
3. If one such limit is flat, then near x_0 the free boundary is a smooth hypersurface, $\{u > 0\}$ is one of its sides, u is smooth up to it from that side, and $u = 0$, $\nabla u = N$ on it [7, 8, 10, 18]. Equivalently, $u = \max\{F, 0\}$ near x_0 for a smooth F with $|\nabla F| = 1$ on $\{F = 0\}$ and $\Delta F = 0$ on $\{F > 0\}$ (extend u across the free boundary by Seeley's theorem [19]).

2.2 Dimension reduction

By item 3, a free boundary point is smooth as soon as *one* blowup there is flat. So to show that a cone u in $\mathbb{R}^n$ is smooth away from its vertex, it suffices to show that its blowups at points $x_0 \neq 0$ are flat. The key observation is that such blowups are effectively lower-dimensional, so that the inductive hypothesis applies to them.

Let $x_0 \neq 0$ be a free boundary point of the cone u. Homogeneity gives an exact identity between rescalings at x_0 at two nearby scales and a translate along x_0; passing to the limit, every blowup U at x_0 is invariant under translation in the direction x_0. So U depends only on the $n-1$ coordinates orthogonal to x_0: $U = u_1 \circ \pi$, where π is the orthogonal projection onto $x_0^{\perp}$. To use the inductive hypothesis on u_1 we need to know that u_1 is itself a minimizer.

Lemma 2.1 (Factors)**.** *If u_1 is a function on $\mathbb{R}^{n-1}$ and $(x', x_n) \mapsto u_1(x')$ is a global minimizer on $\mathbb{R}^n$, then u_1 is a global minimizer on $\mathbb{R}^{n-1}$.*

Idea. A competitor that improves u_1 by a fixed amount on a ball $B \subset \mathbb{R}^{n-1}$ is extended to $B \times [-L, L]$ and cut off linearly in x_n on two unit intervals. The gain is proportional to L, while the cost of the two transition regions is bounded, so for L large the competitor beats $u_1 \circ \pi$. □

So u_1 is a nonzero, one-homogeneous global minimizer in $\mathbb{R}^{n-1}$. If all such minimizers are flat, then u_1 is a half-space solution, hence so is U, and by item 3 the point x_0 is regular. Thus, under the inductive hypothesis, every free boundary point of u other than the origin is regular, with a smooth one-sided chart. (The same mechanism, applied to blowups at singular points of an arbitrary minimizer, is Weiss's dimension reduction behind Corollary 1.2.)

Regularity alone is not enough: the half-space solution is a smooth cone too. The induction needs a component of the positive set on which the cone is genuinely curved, that is, on which its Euclidean Hessian $\nabla^2 u$ does not vanish identically.

Definition 2.2 (Cones with an isolated singularity)**.** A *nonflat minimizing cone with an isolated singularity* in $\mathbb{R}^n$ is a pair (u, D), where u is a continuous, nonnegative, one-homogeneous global minimizer in $\mathbb{R}^n$ and D is a connected component of $\{u > 0\}$, such that every point of $\overline{D} \setminus \{0\}$ has a smooth one-sided chart as in item 3 above with positive side contained in D, and $\nabla^2 u \not\equiv 0$ on D.

Proposition 2.3 (Reduction)**.** *Let $n \geq 2$, and suppose that every nonzero minimizing cone in $\mathbb{R}^{n-1}$ is flat. If u is a nonzero nonflat minimizing cone in $\mathbb{R}^n$, then (u, D) is a nonflat minimizing cone with an isolated singularity for a suitable component D.*

Idea. Charts at points of $\overline{D} \setminus \{0\}$ come from the previous paragraph; the positive side of a chart is connected and so lies in a single component, which gives local separation of components. If $\nabla^2 u$ vanished on every component, each component would be a half-space on which u is linear. Two disjoint half-spaces through the origin would share boundary points other than the origin, contradicting local separation; so there is a single half-space component, and the Bernoulli condition forces the slope to be one. Then u is flat. □

For a cone with an isolated singularity, write $u(ry) = rg(y)$ for $y \in S^m$, and let $\Omega = D \cap S^m$ be the *link*. It is a domain with smooth boundary, and the polar form of the Laplacian and the Bernoulli condition give the overdetermined problem

$$\Delta g = -mg, \quad g > 0 \quad \text{in } \Omega, \qquad g = 0, \quad \nabla g = N \quad \text{on } \partial\Omega. \tag{2.1}$$

At ry the Euclidean Hessian $\nabla^2 u$ vanishes in the radial direction and, on the tangent space of S^m, equals $r^{-1}T(y)$, where

$$T = \nabla^2 g + g\,\mathrm{Id}$$

is the *spherical Hessian of the cone*. Nonflatness means $T \not\equiv 0$ on Ω. The rest of the proof shows that no such (Ω, g) exists for $n \le 6$.

3 Stability: what has to be produced

This section turns minimality into a usable inequality on the link and identifies precisely what object would contradict it. The answer, a positive Jacobi subsolution or more generally a test function paired with a vector field, is the target of everything that follows. Throughout, (u, D) is a nonflat minimizing cone with an isolated singularity in $\mathbb{R}^n$, with link $\Omega \subset S^m$.

3.1 The second variation

Minimality says that every admissible perturbation increases the energy. The natural perturbations are smooth deformations of u near the free boundary. Let φ be a smooth function on $\overline{D}$ with compact support away from the origin. Extend u across ∂D by the smooth function F of a chart and compare u with $(F + \tau\varphi)_+$ for small τ of either sign. The energy changes in two ways: on the fixed domain D the Dirichlet energy changes by $2\tau \int_D \nabla u \cdot \nabla\varphi + \tau^2 \int_D |\nabla\varphi|^2$, and near ∂D the positive set moves. Expanding the moving-boundary contribution in a boundary layer, the linear terms cancel by the Bernoulli condition $|\nabla u| = 1$, the terms involving the normal derivative of φ cancel as well, and what remains is the curvature of the free boundary:

$$J(u_\tau) - J(u) = \tau^2 \left(\int_D |\nabla\varphi|^2 \,\mathrm{d}x - \int_{\partial D} H_D \varphi^2 \right) + o(\tau^2), \qquad H_D = -\nabla^2 u(N, N),$$

where H_D is the mean curvature of the free boundary. Minimality gives the *stability inequality* $\int_{\partial D} H_D \varphi^2 \le \int_D |\nabla\varphi|^2$ for all such φ, with arbitrary boundary values.

3.2 Separation of variables and the Hardy constant

Since the cone is invariant under dilations, it is natural to test with separated functions and to transfer the inequality to the link. Take $\varphi(ry) = \zeta(r)v(y)$. On the cone, $H_D(ry) = r^{-1}H(y)$

with $H = -T(N,N)$ the mean curvature of $\partial\Omega$ in S^m, the volume element is $r^m\,\mathrm{d}r\,\mathrm{d}V$ and the boundary element is $r^{m-1}\,\mathrm{d}r\,\mathrm{d}\sigma$. The stability inequality separates into

$$\left(\int_0^\infty r^{m-2}\zeta^2\right)\int_{\partial\Omega} Hv^2 \le \left(\int_0^\infty r^{m-2}\zeta^2\right)\int_\Omega |\nabla v|^2 + \left(\int_0^\infty r^m\zeta'^2\right)\int_\Omega v^2.$$

The ratio $\int r^m\zeta'^2/\int r^{m-2}\zeta^2$ can be brought arbitrarily close to the sharp radial Hardy constant $((m-1)/2)^2$, using $\zeta(r) = r^{-(m-1)/2}\psi((\log r)/L)$ for a fixed bump ψ and letting $L\to\infty$. With $\beta_m = (m-1)/2$ we obtain the *spherical stability inequality*

$$\int_{\partial\Omega} Hv^2 \le \int_\Omega (|\nabla v|^2 + \beta_m^2 v^2) \tag{3.1}$$

for all smooth v on $\overline{\Omega}$. For links in S^5, $\beta_5^2 = 4$. The Hardy constant is the first place where the dimension enters: it grows quadratically in m.

3.3 Jacobi subsolutions and fluxes

Question 3.1. *What would contradict (3.1)?*

Integrating by parts with the inward normal,

$$\int_\Omega (|\nabla h|^2 + \beta_m^2 h^2) - \int_{\partial\Omega} Hh^2 = -\int_\Omega h(\Delta h - \beta_m^2 h) - \int_{\partial\Omega} h(\partial_N h + Hh). \tag{3.2}$$

So the cone is unstable as soon as there is a nonnegative h, not identically zero, with

$$\Delta h \ge \beta_m^2 h \text{ in } \Omega, \qquad \partial_N h + Hh \ge 0 \text{ on } \partial\Omega, \tag{3.3}$$

strictly somewhere: a positive *Jacobi subsolution*.

The boundary condition in (3.3) ties the flux through $\partial\Omega$ rigidly to h: it involves $\partial_N h$, which is determined by h. We shall see in Section 6 that this boundary condition is exactly where the simplest candidates fail. It is therefore worth separating the two roles of h. The identity (3.2) uses the field $h\nabla h$, for which $\operatorname{div}(h\nabla h) - |\nabla h|^2 - \beta_m^2 h^2 = h(\Delta h - \beta_m^2 h)$; replacing $h\nabla h$ by an arbitrary field gives the following form of the stability inequality.

Lemma 3.2 (Stability in flux form)**.** *Let v be a smooth function and Y a smooth vector field on $\overline{\Omega}$ with $N\cdot Y + Hv^2 \ge 0$ on $\partial\Omega$. Then*

$$\int_\Omega \operatorname{div} Y \le \int_\Omega (|\nabla v|^2 + \beta_m^2 v^2).$$

Proof. By the divergence theorem (the outward normal is $-N$) and the boundary condition, $\int_\Omega \operatorname{div} Y = -\int_{\partial\Omega} N\cdot Y \le \int_{\partial\Omega} Hv^2$, and (3.1) bounds the right side. □

Consequently a cone is unstable as soon as there are a function h and a field Z on $\overline{\Omega}$ with

$$\operatorname{div} Z \ge |\nabla h|^2 + \beta_m^2 h^2 \text{ in } \Omega, \qquad N\cdot Z + Hh^2 \ge 0 \text{ on } \partial\Omega, \tag{3.4}$$

the first inequality strict on a set of positive measure: apply Lemma 3.2 with $v = h$ and $Y = Z$. Now the boundary flux $N\cdot Z$ is free. The difficulty is plain: one must produce such a pair (h, Z) for an *unknown* cone, about which one knows only that it solves (2.1) and is nonflat. (In practice h and Z will be singular where T vanishes, and Lemma 3.2 will be applied to cut-off versions; see Section 9.)

4 How to look for h: calibrations in jet space

Since the cone is unknown, h and Z cannot be written down by hand; they will have to be built by a universal formula from the local behavior of the cone. To see why this is a reasonable hope, and what it leads to, we first look at a one-dimensional example in which the same strategy proves a classical inequality.

Example 4.1 (A Riccati proof of Wirtinger's inequality)**.** We prove $\int_0^L y'^2 \geq \mu \int_0^L y^2$ for $y(0) = y(L) = 0$ and $\mu < (\pi/L)^2$. For a function $F(x, y)$, the derivative of $F(x, y(x))$ is $F_x + F_y\, y'$: it depends on the point $(x, y(x))$ and, linearly, on the value $\eta = y'(x)$, about which the point says nothing. So if

$$\eta^2 - \mu y^2 \geq F_x(x, y) + F_y(x, y)\, \eta \quad \text{for all } (x, y, \eta), \tag{4.1}$$

and $F(x, 0) = 0$, integrating along $x \mapsto (x, y(x), y'(x))$ gives the inequality. The integrand is quadratic, so try $F = \rho(x)y^2$. Then (4.1) reads

$$(\eta - \rho y)^2 - (\rho' + \rho^2 + \mu)y^2 \geq 0 \quad \text{for all } y, \eta,$$

which holds if and only if $\rho' + \rho^2 + \mu \leq 0$: a Riccati inequality. For $\mu = \omega^2$ with $\omega L < \pi$, the function $\rho = \omega \cot(\omega(x + x_0))$, with small $x_0 > 0$, solves it with equality and is smooth on $[0, L]$.

In this example (x, y) is the *base*: the data on which the unknown F is allowed to depend. The derivative $\eta = y'$ is the *fiber*: it appears only after differentiating, it is not determined by the base, so the pointwise inequality must hold for all of its values; and differentiation is linear in it. A function on the base whose derivative along every jet bounds the integrand, for every value of the fiber, is what we call a *calibration in jet space*.

The Bernoulli problem differs in direction. Wirtinger's inequality is a *stability* statement, proved for all y. For cones we want *instability*: h and Z are built from the jet of the cone itself, and the pointwise inequality (3.4), integrated along that one jet, makes the stability form negative. The mechanism is the same: functions on a base whose derivatives are controlled for every value of the free fiber.

Two further features of the example will recur.

Linearity. Condition (4.1) is linear in the unknown F. Finding a calibration is therefore a convex problem, which is what will make a computer search possible (Section 8).

Threshold and obstruction. For $\mu > (\pi/L)^2$ no admissible F exists, because integrating (4.1) along $y_* = \sin(\pi x/L)$ would give $\int (y_*'^2 - \mu y_*^2) \geq 0$, which is false. Actual functions violating the inequality obstruct every calibration. In the Bernoulli problem the De Silva–Jerison cone in $\mathbb{R}^7$ will play the role of y_* (Section 13).

To carry this out for cones we must decide what the base and the fiber are, and what shape h and Z should have. The next section makes these choices.

5 The ansatz

We now decide what h is built from and what form it takes. Two principles guide the choices: the test must see the nonflatness of the cone, and it must respect the scaling of the problem.

Identities satisfied by cones are not needed yet; they enter in Section 6, when a concrete choice is tested.

5.1 Which quantity should h be built from?

Suppose h is given by a universal formula in the jet of the cone, and the pointwise inequalities (3.4) hold, strictly, at every jet where $h \neq 0$. Apply this to the half-space solution, whose link is a hemisphere with $T \equiv 0$: if h did not vanish at such jets, integration would show that the half-space solution is unstable, which is absurd. So h must vanish where $T = 0$, and the natural quantity to build it from is the spherical Hessian T of Section 2 and its size

$$w = |T|.$$

This is the Bernoulli counterpart of Simons's use of the second fundamental form for minimal cones.

5.2 Scaling, base and fiber

The inequalities (3.4) are homogeneous in h and Z, and multiplying the cone by a constant multiplies T by the same constant. It is therefore natural to separate the size w of the Hessian from its shape

$$A = T/w,$$

which is a unit symmetric trace-free matrix wherever $w > 0$ (trace-free because $\operatorname{tr} T = \Delta g + mg = 0$ by (2.1)). Besides A, the lower-order data available at a point are the values $p = \nabla g$ and g. We take these as the base and let h be a power of w times an unknown function of the base:

$$h = w^a \sqrt{f(A, p, g)}, \qquad a > 0, \quad f > 0. \tag{5.1}$$

The square root is there because the inequalities involve $h^2 = w^{2a} f$, which is then linear in the unknown f; as in Example 4.1, linearity in the unknowns keeps the search convex. The fiber consists of what appears only after differentiating: the third derivatives of g (that is, ∇T), and w itself, which has been scaled out of the base. The flux Z will be chosen in the same spirit. Its simplest choice is a multiple of $h\nabla h$, as in (3.2), and the simplest instance of (5.1) is $f = 1$. We try this first.

6 The simplest instance: plain powers

The first case of the ansatz is $f = 1$, so $h = w^a$, the analogue of Simons's test function. Testing it will tell us which properties of cones the method needs, and exactly where the dimension enters.

Question 6.1. *Does $h = w^a$, with Z a multiple of $h\nabla h$, satisfy (3.4)?*

Fix $a > 0$ and $c > 0$ and set

$$h = w^a, \qquad Z = (c/a) h \nabla h = c w^{2a-1} \nabla w.$$

For $c = a$ this is $Z = h\nabla h$ and (3.4) is the subsolution property (3.3). Since we need strict inequality somewhere, and since $h^2 = w^{2a}$, the natural form of the requirement is

$$\operatorname{div} Z - |\nabla h|^2 - \beta_m^2 h^2 \geq \varepsilon w^{2a} \tag{6.1}$$

for some $\varepsilon > 0$. Computing $\operatorname{div} Z = cw^{2a-2}(w\Delta w + (2a-1)|\nabla w|^2)$ requires Δw: a second derivative of T, that is, a fourth derivative of g. Before going further we need to know what the equation says about it.

6.1 What the equation says about ΔT

Proposition 6.2 (Bochner identity)**.** *On Ω, T is symmetric and trace-free, ∇T is fully symmetric and trace-free, and*

$$\Delta T = mT, \qquad \Delta w^2 = 2mw^2 + 2|\nabla T|^2.$$

Trace-freeness is the equation $\Delta g = -mg$. Full symmetry of ∇T is the Codazzi property of T: on the round sphere the curvature terms in $\nabla_i\nabla_j\nabla_k g - \nabla_j\nabla_i\nabla_k g$ cancel against the derivatives of $g\,\mathrm{Id}$. Then $\Delta T_{jk} = \nabla_i\nabla_j T_{ik}$, and commuting ∇_i past ∇_j produces curvature terms of S^m which combine with $\operatorname{tr} T = 0$ to give mT. The second identity is the product rule, $\Delta\,|T|^2 = 2\langle T, \Delta T\rangle + 2|\nabla T|^2$; it is the Bernoulli analogue of Simons's identity.

Normalize the fiber as we did the base: where $w > 0$ put

$$X = \nabla T / w \in \mathrm{STF}^3(\mathbb{R}^m), \qquad q = \nabla \log w, \quad \text{so that } q_l = X_{ljk}A_{jk},$$

and $|\nabla T|^2 = w^2\ |X|^2$. Then $w\Delta w = mw^2 + |\nabla T|^2 - |\nabla w|^2$ gives

$$w^{-2a}(\operatorname{div} Z - |\nabla h|^2 - \beta_m^2 h^2) = \underbrace{mc - \beta_m^2}_{\text{base only}} + \underbrace{c\ |X|^2 - \big((2-2a)c + a^2\big)\ |q|^2}_{\text{quadratic in the fiber } X}. \tag{6.2}$$

As in Example 4.1, this must be at least ε for all fiber values $X \in \mathrm{STF}^3$, since nothing we know about cones restricts ∇T beyond symmetry and trace-freeness. Bounding $|q|$ in terms of $|X|$ is an algebraic form of the improved Kato inequality; on $\mathbb{R}^2$ one has exactly $|q|^2 = |X|^2\ /2$.

6.2 What happens at the boundary

The boundary inequality involves $N \cdot Z + Hh^2 = cw^{2a-1}\partial_N w + Hw^{2a}$, so we need to know H, $\partial_N w$, and that w does not vanish there. The overdetermined conditions $g = 0$, $\nabla g = N$ give this through a maximum principle argument applied to $|\nabla u|^2 = g^2 + |p|^2$.

Lemma 6.3 (Gradient bound and boundary signs)**.** *On the link of a cone with an isolated singularity, $\chi := 1 - g^2 - |p|^2 > 0$ in Ω, and on $\partial\Omega$ one has $H = -T(N,N) > 0$, N is an eigenvector of T, and $w > 0$. In particular the zero set of w is a compact subset of Ω.*

Indeed $g^2 + |p|^2$ satisfies $\Delta(g^2 + |p|^2) = 2w^2 \geq 0$ and $\nabla(g^2 + |p|^2) = 2Tp$; it equals one on $\partial\Omega$ and is not constant since $T \not\equiv 0$. The strong maximum principle and Hopf's lemma give $\chi > 0$ inside and $2T(N,N) = \partial_N(g^2 + |p|^2) < 0$ on $\partial\Omega$, and tangential differentiation of $g^2 + |p|^2 = 1$ gives $T(N,\tau) = 0$ for tangent vectors τ.

Let $\kappa = H/w > 0$ be the *normalized mean curvature*; then $AN = -\kappa N$. Differentiating $T(N,\tau) = 0$ and the trace condition along $\partial\Omega$ shows that every component of ∇T with an odd number of normal slots is determined by T:

$$\nabla_N T_{\alpha\beta} = -w^2(\kappa A + A^2)_{\alpha\beta}, \quad \nabla_N T_{NN} = w^2, \tag{6.3}$$

for tangential α, β, while the components with an even number of normal slots involve tangential derivatives of the curvature of $\partial\Omega$ and are not determined. Since T has no mixed normal-tangential components, $\partial_N w = T_{jk}\nabla_N T_{jk}/w$ only involves the determined ones:

$$\partial_N w = -w^2(2\kappa + \operatorname{tr} A^3). \tag{6.4}$$

Hence

$$w^{-2a-1}(N \cdot Z + Hh^2) = \kappa(1-2c) - c \operatorname{tr} A^3. \tag{6.5}$$

At the boundary no fiber is left: this depends only on the *boundary shape* of A, that is, on $\kappa > 0$ and the $m-1$ tangential eigenvalues $x_1, ..., x_{m-1}$, with $\sum x_i = \kappa$ and $\sum x_i^2 + \kappa^2 = 1$. Nothing we know about cones excludes any such shape, so a universal formula must handle all of them.

6.3 The window, dimension by dimension

Example 6.4 (Links in S^2: the window is open)**.** For $m = 2$, $\beta_2^2 = 1/4$ and a unit trace-free 2×2 matrix has eigenvalues $\pm 1/\sqrt{2}$. Hence $\operatorname{tr} A^3 = 0$, $\kappa = 1/\sqrt{2}$, and (6.5) equals $(1 - 2c)/\sqrt{2}$. The interior form (6.2) is $2c - 1/4 + (ac - a^2/2)|X|^2$, using $|q|^2 = |X|^2/2$. So both inequalities hold, with a positive margin in the interior, whenever

$$\max\left\{\frac{1}{8}, \frac{a}{2}\right\} < c < \frac{1}{2}.$$

This is nonempty for every $a < 1$. With $a = 29/50$ and $c = 2/5$ one gets precisely the certificate used to exclude nonflat cones in $\mathbb{R}^3$ (Section 10).

Example 6.5 (Links in S^3: the window closes)**.** For $m = 3$, $\beta_3^2 = 1$. Taking $X = 0$ in (6.2) shows that a positive interior margin requires $c > 1/3$. At the boundary, $\kappa \in \left(0, \sqrt{2/3}\right]$, and the tangential eigenvalues satisfy $x_1 + x_2 = \kappa$ and $x_1 x_2 = \kappa^2 - 1/2$, so $\operatorname{tr} A^3 = (3/2)\kappa(1 - 2\kappa^2)$ and

$$\kappa(1-2c) - c \operatorname{tr} A^3 = \kappa(1 - 7c/2 + 3c\kappa^2).$$

For small κ this is negative unless $c \le 2/7$. Since $2/7 < 1/3$, no choice of a and c works. The bad shapes have small κ, that is, boundary mean curvature small compared with the Hessian, with principal curvatures close to $\pm w/\sqrt{2}$.

Example 6.6 (Links in S^m, $m \ge 4$)**.** Now the interior needs $c > \beta_m^2/m = (m-1)^2/(4m)$, which is $9/16$ for $m = 4$ and $4/5$ for $m = 5$. The boundary fails for every $c > 0$: as $\kappa \to 0$ the tangential eigenvalues can approach any unit trace-free configuration of $m - 1 \ge 3$ numbers, for instance $(2, -1, -1, 0, ...)/\sqrt{6}$, for which $\operatorname{tr} A^3 > 0$, and then (6.5) tends to $-c \operatorname{tr} A^3 < 0$.

These examples are the concrete form of the incompatibility described by Jerison and Savin [5]: the interior wants a large coefficient to beat the Hardy constant, while the boundary wants a small one, and their worst cases come from different Hessian shapes. The failure is pointwise, at particular shapes, and at the boundary it is exactly the rigid link between $N \cdot Z$ and h that Lemma 3.2 was designed to loosen.

7 The general ansatz

This section returns to the ansatz of Section 5 and enriches it exactly where the plain power fails, and then turns (3.4) into polynomial inequalities in finitely many variables.

Question 7.1. *What extra freedom could repair the plain power?*

Two kinds. First, the weight f in (5.1), which can be small on bad boundary shapes and large elsewhere. Second, a flux Z that is no longer a multiple of $\nabla(h^2)$, so that $N \cdot Z$ decouples from $\partial_N h$ and can be adjusted on the bad shapes. (The certificate on $\mathbb{R}^3$ in Section 10 uses only the second freedom: $f = 1$, with extra flux terms. This is where the real gain lies.)

7.1 Letting the weight see p and g

The interior and the boundary differ in the values of (p, g): by Lemma 6.3, $\chi > 0$ in Ω, while $g = 0$ and $|p| = 1$ on $\partial\Omega$. A weight $f(A, p, g)$ can therefore behave differently at the boundary and in the interior. The base is the compact set

$$\mathcal{K}_m = \{(A, p, g) : A \in \mathrm{STF}^2(\mathbb{R}^m), |A| = 1, \chi = 1 - |p|^2 - g^2 \geq 0\},$$

and the gradient bound enters the algebra as the single constraint $\chi \geq 0$.

To differentiate functions on the base along the cone we use the chain rule and

$$\nabla_l A_{ij} = X_{lij} - q_l A_{ij}, \quad \nabla_l p_i = w A_{li} - g\delta_{li}, \quad \nabla_l g = p_l, \quad \nabla_l w = w q_l, \tag{7.1}$$

which follow from the definitions. For a function $\varphi(A, p, g)$, $\nabla_l \varphi$ is thus a polynomial in the base and fiber variables, linear in the fiber, exactly as $F_x + F_y \eta$ was in Example 4.1. From now on we treat w and X as free variables in this expression; w enters only linearly, so forgetting that $w = |T|$ is a harmless relaxation.

7.2 The general flux

Question 7.2. *What is the most general flux built from the jet that we can hope to control?*

Z should have the same homogeneity as $h\nabla h$, namely w^{2a} times terms of degree at most one in the fiber variables X and w. So

$$Z_l = w^{2a}\big(\Gamma_{lijk} X_{ijk} + w M_l + P_l\big),$$

with coefficients Γ, M, P depending on the base and equivariant under rotations. For the plain power, $\Gamma_{lijk} = c\delta_{li} A_{jk}$ and $M = P = 0$. But $\operatorname{div} Z$ now contains ∇X, that is, all fourth derivatives of g, and we must ask what the equation says about them.

A lower-order model shows what to expect. In the plane, let u be harmonic, and consider a field $Y(u, \nabla u)$ built from the one-jet. Then $\operatorname{div} Y = \partial_u Y_i \, \partial_i u + \partial_{p_j} Y_i \, \partial_i \partial_j u$. The trace-free part of the Hessian is not determined by the one-jet and enters linearly, so a lower bound independent of it requires the symmetric part of $\partial_p Y$ to be a multiple of the identity; the antisymmetric part of $\partial_p Y$ pairs to zero with the symmetric Hessian and is unrestricted.

For cones, at a point $\nabla_l \nabla_i T_{jk}$ is symmetric in (i, j, k), and beyond that only two combinations are determined: the trace in (l, i), which is $\Delta T = mT$ by Proposition 6.2, and the antisymmetric part in (l, i), which is the curvature of the sphere acting on T:

$$\nabla_l \nabla_i T_{jk} - \nabla_i \nabla_l T_{jk} = \delta_{lj} T_{ik} + \delta_{lk} T_{ij} - \delta_{ij} T_{lk} - \delta_{ik} T_{lj}. \tag{7.2}$$

The remaining, symmetric trace-free, part is free and would enter $\operatorname{div} Z$ linearly; a term linear in a free variable destroys any lower bound. So, as in the planar model, the coefficient Γ should see only the trace and the antisymmetric part: after symmetrizing in (i,j,k), which does not change $\Gamma_{lijk}X_{ijk}$, we may take

$$\Gamma_{lijk} = \delta_{li}U_{jk} + C_{lijk}, \qquad C_{lijk} = -C_{iljk},$$

with unknown tensors U and C. Then

$$Z_l = w^{2a}\big(U_{jk}X_{ljk} + C_{lijk}X_{ijk} + wM_l + P_l\big), \tag{7.3}$$

and the fourth derivatives enter $\operatorname{div} Z$ only through

$$\big(U_{jk}\nabla_l\nabla_l T_{jk} + C_{lijk}\nabla_l\nabla_i T_{jk}\big)/w = mU_{jk}A_{jk} + C_{lijk}\big(\delta_{lj}A_{ik} + \delta_{lk}A_{ij}\big), \tag{7.4}$$

using $\Delta T = mT$ and, since C is antisymmetric, $C_{lijk}\nabla_l\nabla_i T_{jk} = (1/2)C_{lijk}(\nabla_l\nabla_i - \nabla_i\nabla_l)T_{jk}$.

The boundary raises the same question one order lower. There $N\cdot Z$ contains $p_lU_{jk}X_{ljk} + p_lC_{lijk}X_{ijk}$, and the even-normal components of ∇T are undetermined by (6.3). If U and C are even functions of p, then at a boundary point the three-tensor $p_l\Gamma_{lijk}$ is odd under the reflection in $p^\perp$, which fixes A (as p is an eigenvector) and reverses p. A tensor odd under this reflection has no components with an even number of normal slots, so the undetermined part drops out. Finally, $p\cdot P$ would contribute to $N\cdot Z$ at order w^{2a}, below the order w^{2a+1} of every other boundary term; to keep the boundary inequality homogeneous in w we require $P=0$ when $g=0$. To summarize: the unknowns are f, U, C, M, P, functions on the base, equivariant under rotations, with U and C even in p, C antisymmetric in its first two slots, and $P=0$ at $g=0$.

7.3 From the target inequalities to two polynomials

Question 7.3. *How do we turn (3.4) for this ansatz into something checkable?*

The divergence. By (7.1), $\nabla_l w^{2a} = 2aw^{2a}q_l$ and $\nabla_l X_{ijk} = \nabla_l\nabla_i T_{jk}/w - q_lX_{ijk}$. Applying the product rule to (7.3) and using (7.4),

$$\begin{aligned} w^{-2a}\operatorname{div} Z = {} & mU_{jk}A_{jk} + C_{lijk}\big(\delta_{lj}A_{ik} + \delta_{lk}A_{ij}\big) + \big(\nabla_l U_{jk} + (2a-1)q_lU_{jk}\big)X_{ljk} \\ & + \big(\nabla_l C_{lijk} + (2a-1)q_lC_{lijk}\big)X_{ijk} + w(\nabla_l M_l + (2a+1)q_lM_l) + \nabla_l P_l + 2aq_lP_l, \end{aligned}$$

where ∇_l of a coefficient is computed by the chain rule from (7.1). This is a polynomial in the base and fiber variables.

The test function. $|\nabla h|^2 = h^2\,|\nabla\log h|^2$ is rational in f, since $\nabla\log h = aq + \nabla f/(2f)$. A free vector ℓ linearizes it, exactly as the square $(\eta-\rho y)^2$ did in Example 4.1: for every ℓ,

$$f\,|\ell|^2 - 2f\ell\cdot\nabla\log h = f\,|\ell - \nabla\log h|^2 - f\,|\nabla\log h|^2,$$

with equality at $\ell = \nabla\log h$, and $2f\nabla\log h = \nabla f + 2aqf$.

The margin. As in (6.1) we ask for εw^{2a}, the natural size of $h^2 = w^{2a}f$. The margin is needed twice: for the contradiction, and to absorb the errors of the cutoffs in Section 9.

Before stating the result, here are all the variables that occur, with the only information about the cone that will be used.

variable	geometric value	constraint used
A	T/w	$A \in \mathrm{STF}^2(\mathbb{R}^m)$, $\lvert A\rvert = 1$
p, g	∇g, g	$\chi = 1 - \lvert p\rvert^2 - g^2 \geq 0$
X	$\nabla T/w$	$X \in \mathrm{STF}^3(\mathbb{R}^m)$, otherwise free
w	$\lvert T\rvert$	free
ℓ	$\nabla \log h$	free
q	$\nabla \log w$	derived: $q_l = X_{ljk}A_{jk}$

Definition 7.4 (Interior and boundary polynomials)**.** The *interior polynomial* $\mathcal{I}_m$ is the right side of the formula for $w^{-2a}\operatorname{div} Z$ above, minus $\beta_m^2 f$, plus $f\ |\ell|^2 - (\nabla_l f + 2aq_l f)\ell_l$. The *boundary polynomial* is

$$\mathcal{B}_m = \kappa f + p_l U_{jk} V_{ljk} + p_l C_{lijk} V_{ijk} + p_l M_l,$$

evaluated at $g = 0$, $|p| = 1$, $Ap = -\kappa p$, where V is the three-tensor whose components with an odd number of slots equal to p are those of $\nabla T/w^2$ given by (6.3), and whose other components vanish.

By construction one has the exact identities

$$\begin{aligned} \mathcal{I}_m &= w^{-2a}(\operatorname{div} Z - \beta_m^2 h^2 - |\nabla h|^2) + f\ |\ell - \nabla \log h|^2, \\ N \cdot Z + Hh^2 &= w^{2a+1}\mathcal{B}_m. \end{aligned} \tag{7.5}$$

For the plain power, $\mathcal{I}_m$ minimized over ℓ is (6.2), and $\mathcal{B}_m$ is (6.5).

7.4 The certificate

Question 7.5. *What exactly must the coefficients satisfy?*

By (7.5), it suffices that $\mathcal{I}_m \geq \varepsilon$ and $\mathcal{B}_m \geq 0$ for every value of the variables that a cone could produce. Since we cannot say more about these values than the table does, we ask for this on the whole table.

Definition 7.6 (Certificate on $\mathbb{R}^m$)**.** Fix $a \in (1/2, 1)$. A *certificate on* $\mathbb{R}^m$ consists of smooth (in practice polynomial) maps f, U, C, M, P on the base with the symmetries above and $f > 0$, and a constant $\varepsilon > 0$, such that

1. $\mathcal{I}_m \geq \varepsilon(1 + |\ell|^2)$ for all $(A, p, g) \in \mathcal{K}_m$, all $X \in \mathrm{STF}^3(\mathbb{R}^m)$, all $w \in \mathbb{R}$ and $\ell \in \mathbb{R}^m$;
2. $\mathcal{B}_m \geq 0$ for all unit trace-free A and unit p with $Ap = -\kappa p$, $\kappa > 0$.

The range of a is explained later: $a > 1/2$ makes h have finite energy near zeros of w (Section 9), and $a < 1$ is used for the suspension (Section 11). A certificate yields, wherever $w > 0$, $\operatorname{div} Z \geq |\nabla h|^2 + \beta_m^2 h^2 + \varepsilon w^{2a}$ in Ω (take $\ell = \nabla \log h$) and $N \cdot Z + Hh^2 \geq 0$ on $\partial\Omega$, which is (3.4) with a margin. The definition no longer mentions any cone: it is a statement about polynomials on a compact set of tensors.

Theorem 7.7 (Certificates)**.** *With $a = 29/50$, certificates exist on $\mathbb{R}^2$, on $\mathbb{R}^3$ and on $\mathbb{R}^5$.*

The certificate on $\mathbb{R}^5$ is the main contribution of [1]; those on $\mathbb{R}^2$ and $\mathbb{R}^3$ are much smaller and are described in Section 10. The following table recapitulates the analogy with Example 4.1.

Riccati toy model	**Bernoulli cones**
base (x, y)	normalized two-jet (A, p, g)
fiber $\eta = y'$, free	$X = \nabla T/w$ and w (free), auxiliary ℓ
unknown $F = \rho(x)y^2$	unknown coefficients f, U, C, M, P
derivative $F_x + F_y\eta$	chain rule with (7.1)
(4.1) for every η	$\mathcal{I}_m \geq \varepsilon(1 + \lvert\ell\rvert^2)$ for every (w, ℓ, X)
completing the square in η	completing the square in ℓ
boundary term $F(L, 0) - F(0, 0)$	$\mathcal{B}_m \geq 0$ on boundary shapes
threshold $\omega = \pi/L$	link dimension five versus six
obstruction $\sin(\pi x/L)$	the De Silva–Jerison cone in $\mathbb{R}^7$

Table 1: A dictionary between the toy calibration and the Bernoulli certificate.

8 Finding and verifying a certificate

This section explains why the existence of a certificate is a question that a computer can settle, how one searches for one, and what the certificate on $\mathbb{R}^5$ looks like. It can be skipped on a first reading: the rest of the argument uses only Definition 7.6.

Question 8.1. *Why should a finite computation suffice, and why is exact arithmetic the natural tool?*

Everything in Definition 7.6 is algebraic. The base $\mathcal{K}_m$ is compact, and the unknowns f, U, C, M, P enter *linearly*: $\mathcal{I}_m$ is linear in them and their derivatives, and $\mathcal{B}_m$ in their values. One more step puts the interior condition in a standard form. $\mathcal{I}_m$ has terms of degree zero, one and two in the fiber variables (w, ℓ, X). Introducing an auxiliary variable z and multiplying each term by the power of z that brings its degree to two turns $\mathcal{I}_m$ into a quadratic form in (w, z, ℓ, X) whose coefficients are polynomials on the base, and by scaling, condition (1) is equivalent to this form being at least $\varepsilon(z^2 + \lvert\ell\rvert^2)$. So, as in the Riccati example, the existence of a certificate is a convex feasibility problem: find coefficients such that a family of quadratic forms, parametrized by a compact semialgebraic set, is uniformly positive in the relevant directions. Two consequences shape the proof.

First, the problem can be attacked by semidefinite programming. Choose a finite basis of equivariant polynomial coefficients (contractions of A, p, g and the identity), and look for nonnegativity proofs in the form of sums of squares: Gram matrices that exhibit the quadratic form, up to a controlled remainder, as a sum of squares of polynomials, possibly multiplied by the nonnegative function χ that encodes the gradient bound [20]. The unknown coefficients and the Gram matrices enter linearly, so both can be searched for at once. This is how certificates of this kind are found numerically.

Second, a numerical solution is not a proof, but a strict one is robust. Because the desired inequalities have a positive margin, one can round the numerical output to rational numbers, write the inequality as an *exact* identity: the quadratic form equals a sum of manifestly nonnegative Gram terms, plus reserved positive multiples of $w^2, z^2, \lvert\ell\rvert^2, \lvert X\rvert^2$, minus a remainder with small rational coefficients. The remainder is then bounded crudely, term by term, using only elementary inequalities valid on the compact base. All the arithmetic is with integers and rationals, and the result is a finite list of checks.

8.1 The certificate on $\mathbb{R}^5$

Having seen why a certificate is a finite object, one may ask what the one on $\mathbb{R}^5$ actually consists of. We describe its structure without its numbers; each ingredient corresponds to one of the steps just outlined.

The ansatz. The coefficients f, U, C, M, P are rational linear combinations of a few hundred tensor monomials in A, p, g and $\operatorname{tr} A^3$: contracted products of powers of A and copies of p. Equivariance is automatic for such contractions, and parity in p is visible from the number of copies of p.

The interior. The nonnegative part consists of blocks, each a sum of squared norms of tensors that are linear in the fiber, possibly after projecting onto the antisymmetric, the symmetric trace-free, or a mixed symmetry type, and possibly weighted by χ. After subtracting these blocks and the reserved diagonal terms, the remainder is a sum of scalar contractions of degree two in the fiber. Each is bounded by elementary estimates on the base, such as $\|A\|_{\mathrm{op}}^2 \le 4/5$, $|\operatorname{tr} A^3| \le 3/\sqrt{20}$, and the improved Kato bound $|q|^2 \le (5/7)|X|^2$, and a weighted Young inequality shows that the reserved terms absorb the total with room to spare. (The bound on $\operatorname{tr} A^3$ has a one-line proof: with $c_* = \sqrt{4/5}$ bounding the eigenvalues α_i, one has $\sum_i (\alpha_i - c_*)(\alpha_i + 1/(5c_*))^2 \le 0$, which expands to $\operatorname{tr} A^3 \le c_* - 1/(5c_*) = 3/\sqrt{20}$.)

The boundary. By equivariance one may take p to be the last basis vector and A diagonal, with tangential eigenvalues $x_1, \dots, x_4$. Then $\mathcal{B}_5$ becomes a symmetric polynomial in $x_1, \dots, x_4$ on the set $\sum x_i^2 + (\sum x_i)^2 = 1$, $\sum x_i = \kappa > 0$. It is written as a sum of squares, one block weighted by κ, plus a positive constant, minus a remainder whose finitely many coefficients are summed in absolute value. This gives $\mathcal{B}_5$ bounded below by an explicit positive rational.

Remark 8.2 (What the certificate proves)**.** The certificate is stronger than what the geometry needs: w is treated as a free real variable rather than $|T|$, X as an arbitrary trace-free tensor unrelated to the base, and only $\chi \ge 0$ is used of the strict gradient bound. This is what makes it a finite statement.

9 From pointwise to global: Hessian zeros

A certificate gives the pointwise inequalities only where $w > 0$: the fields h and Z are built from $A = T/w$ and $X = \nabla T/w$, and are singular where the Hessian vanishes. To apply Lemma 3.2 we must integrate across this zero set, about which nothing is known. This section shows that it is invisible to the energy. The argument multiplies by cutoffs and shows that the errors vanish in the limit.

9.1 The weighted Bochner estimate

Question 9.1. *Why must the exponent a be larger than $\frac{1}{2}$?*

Near a zero of w the test function $h \approx w^a$ has $|\nabla h|^2 \approx a^2 w^{2a-2} |\nabla w|^2 \le a^2 w^{2a-2} |\nabla T|^2$. So h has finite energy, and the cutoff errors are integrable, if $w^{2a-2}|\nabla T|^2$ is integrable. Nothing is known about the zero set of w (its size, regularity or dimension), but the Bochner identity of Proposition 6.2, together with $|\nabla w^2|^2 \le 4w^2 |\nabla T|^2$, gives exactly this integrability when $a > 1/2$.

Lemma 9.2 (Weighted Bochner estimate)**.** *Let Ω be the link of a cone with an isolated singularity and $1/2 < a \le 1$. Then $\int_{\Omega\cap\{w>0\}} w^{2a-2}\ |\nabla T|^2 < \infty$.*

Proof. Put $W = w^2 + \gamma$ with $\gamma > 0$, which is smooth. Since $\Delta W \ge 2|\nabla T|^2$ and $|\nabla W|^2 \le 4w^2\ |\nabla T|^2 \le 4W\ |\nabla T|^2$,

$$\Delta(W^a) = aW^{a-1}\Delta W - a(1-a)W^{a-2}\ |\nabla W|^2 \ge a(4a-2)W^{a-1}\ |\nabla T|^2.$$

Multiply by a cutoff ζ that equals one near the zero set of w (which is compact in Ω by Lemma 6.3) and integrate by parts twice: $a(4a-2)\int \zeta W^{a-1}\ |\nabla T|^2 \le \int |\Delta\zeta|\ W^a$, uniformly in γ. Let $\gamma \to 0$; away from the zero set the integrand is bounded. □

9.2 Cutoffs and the contradiction

With integrability in hand, we can run the argument of Lemma 3.2 on the region where $w > 0$, cutting off near the zero set and checking that nothing is lost in the limit. Let ϑ be a smooth profile with $\vartheta = 0$ on $[0,1]$ and $\vartheta = 1$ on $[2,\infty)$, and take $\xi_j = \vartheta(jw)$, which vanishes near the zero set of w. Apply Lemma 3.2 to $v = \xi_j h$ and $Y = \xi_j^2 Z$, whose boundary flux satisfies $N\cdot(\xi_j^2 Z) + H(\xi_j h)^2 = \xi_j^2(N\cdot Z + Hh^2) \ge 0$. After expanding $|\nabla(\xi_j h)|^2$ and $\operatorname{div}(\xi_j^2 Z)$, the terms $\operatorname{div} Z - |\nabla h|^2 - \beta_m^2 h^2$ appear multiplied by ξ_j^2, and the interior inequality gives

$$\varepsilon \int_\Omega \xi_j^2 w^{2a} \le \int_\Omega \big(h^2\ |\nabla\xi_j|^2 + 2\xi_j h\nabla h\cdot\nabla\xi_j - 2\xi_j Z\cdot\nabla\xi_j\big). \tag{9.1}$$

Since $|\nabla w| \le |\nabla T|$, one has $|\nabla\xi_j| \le C\ |\nabla T|/w$ on the transition region, and the bounds $|h| \le Cw^a$, $|\nabla h| + |Z|\ w^{-a} \le Cw^a(1 + w + |\nabla T|/w)$, which follow from the compactness of the base, show that every error is dominated by $Cw^{2a}(1+|\nabla T|/w)^2$, integrable by Lemma 9.2. Each error vanishes pointwise for large j, so the right side of (9.1) tends to zero by dominated convergence, while the left side tends to $\varepsilon\int_\Omega w^{2a} > 0$ because $T \not\equiv 0$. This is a contradiction.

Theorem 9.3 (Exclusion from a certificate)**.** *Let $n \ge 3$. If a certificate on $\mathbb{R}^{n-1}$ exists, there is no nonflat minimizing cone with an isolated singularity in $\mathbb{R}^n$.*

This step uses only $a > 1/2$; the upper bound $a < 1$ enters in Section 11.

10 Low dimensions

Theorem 9.3 reduces the classification in a given dimension to the construction of a certificate. In low dimensions the certificates are small enough to describe completely, and they show the method at work. Each step of the induction uses Proposition 2.3 and an exclusion of cones with an isolated singularity.

Dimension one. A one-homogeneous function on $\mathbb{R}$ is $b_1 x_+ + b_2(-x)_+$. If both slopes are positive, adding a small bump at the origin lowers the energy at first order, since the positive set does not change while the Dirichlet energy decreases. So $u = bx_+$, and comparing with the translates $b(x-t)_+/(1-t)$ on $(-1,1)$, whose energy $b^2/(1-t) + (1-t)$ must be minimal at $t = 0$, forces $b = 1$.

Dimension two. The link lies in S^1, so T is a trace-free 1×1 matrix and vanishes. A cone with an isolated singularity in $\mathbb{R}^2$ would have $\nabla^2 u \equiv 0$ on D, which Definition 2.2 excludes.

Dimension three. The certificate on $\mathbb{R}^2$ is the plain power of Example 6.4: $f = 1$, $U = (2/5)A$, $C = M = P = 0$, with $a = 29/50$. Its verification is a completed square. Since $|X|^2 = 2\,|q|^2$ for trace-free tensors on $\mathbb{R}^2$,

$$\mathcal{I}_2 - \frac{1}{10}(1 + |\ell|^2) = \frac{9}{20} + \frac{9}{10}\,|\ell - \frac{29}{45}q|^2 + \frac{203}{2250}\,|q|^2 \geq 0,$$

and $\mathcal{B}_2 = \kappa/5 = \sqrt{2}/10$. With Theorem 9.3 this excludes nonflat cones in $\mathbb{R}^3$, recovering the theorem of Caffarelli, Jerison and Kenig [4].

Dimension four. By Example 6.5 the plain power fails on $\mathbb{R}^3$, so the certificate must use the extra freedom. A small one suffices, and it uses only the flux: $f = 1$ and $C = P = 0$, with U a combination of five equivariant terms ($p \otimes p$, $p \otimes Ap$, $Ap \otimes Ap$, A and A^2; the coefficient of A depends on $p \cdot Ap$ and $|p|^2$, that of A^2 on $|p|^2$) and M a combination of p and Ap. The flux is no longer a multiple of $\nabla(h^2)$: the terms in p and the vector M give the boundary flux the flexibility that the bad shapes of Example 6.5 require. Both inequalities reduce, by equivariance, to diagonal A. At the boundary the polynomial becomes a quartic in κ on $\left(0, \sqrt{2/3}\right]$, bounded below by an explicit Positivstellensatz identity. In the interior one obtains a quadratic form in twelve fiber coordinates (the seven independent entries of X, and w, z, ℓ) with coefficients polynomial in six base coordinates; it is bounded below by a sum-of-squares identity with multipliers 1 and χ, the constraint $|A| = 1$ entering through a polynomial multiple, and a small residual absorbed by a reserved multiple of the fiber norm. With Theorem 9.3 this excludes nonflat cones in $\mathbb{R}^4$, recovering in this setting the theorem of Jerison and Savin [5].

11 Dimension five: the suspension

Theorem 9.3 with the certificate on $\mathbb{R}^5$ excludes cones in $\mathbb{R}^6$. For cones in $\mathbb{R}^5$ one would need a certificate on $\mathbb{R}^4$, which has not been constructed. This section explains the device that avoids it: a cone in $\mathbb{R}^5$ is turned into a cone in $\mathbb{R}^6$, at the price of two singular points on the link, which must then be removed in the integration.

11.1 The suspended cone

Lemma 11.1 (Products)**.** *If u_1 is a global minimizer on $\mathbb{R}^{n-1}$, then $(x', x_n) \mapsto u_1(x')$ is a global minimizer on $\mathbb{R}^n$.*

Idea. Slice a competitor on a ball of $\mathbb{R}^n$ by the hyperplanes $\{x_n = \text{const}\}$; almost every slice is an admissible competitor for u_1, and the term $|\partial_{x_n} V|^2$ only adds energy. □

Let (u_o, D_o) be a nonflat minimizing cone with an isolated singularity in $\mathbb{R}^5$, with link $\Omega_o \subset S^4$, restriction g_o and Hessian norm w_o. By Lemma 11.1, $\tilde{u}(x', x_6) = u_o(x')$ is a minimizing cone in $\mathbb{R}^6$. Its singular set is the axis $\{0\} \times \mathbb{R}$, which meets S^5 in two points, the *tips*. In coordinates $(t, y) \mapsto (\sin t\, y, \cos t)$, $0 < t < \pi$, $y \in \Omega_o$, the link of $D_o \times \mathbb{R}$ is

$$\Omega = \{(\sin t\, y, \cos t) : y \in \Omega_o\}, \quad \tilde{g} = \sin t\, g_o(y),$$

with the metric $\mathrm{d}t^2$ plus $\sin^2 t$ times the round metric of S^4. It is smooth away from the two tips, and all the local facts of Section 6 and Section 7 hold there with $m = 5$. The stability inequality (3.1) holds for tests supported away from the tips, with $\beta_5^2 = 4$. The gradient bound and boundary signs carry over from Ω_o: $\tilde{g}^2 + |\nabla \tilde{g}|^2 = g_o^2 + |\nabla g_o|^2$, $\tilde{w} = w_o / \sin t$ and

$\tilde{H} = H_o/\sin t$. So a certificate on $\mathbb{R}^5$ gives the pointwise inequalities on the suspended link wherever $\tilde{w} > 0$, that is, away from the tips and the suspended zero set of w_o.

11.2 Why the tips cost nothing

Use the product cutoff $\xi_j = \vartheta(jw_o(y))\vartheta(j\sin t)$, which vanishes near the zero set of w_o and near both tips. On the suspension, $\tilde{w} = w_o/\sin t$, $|\nabla\xi_j| \lesssim (1 + |\nabla T_o|/w_o)/\sin t$, and the volume element is $\sin^4 t\,\mathrm{d}t\,\mathrm{d}V_o$. Each error in (9.1) is therefore bounded by a constant times

$$(\sin t)^{4-2-2a}\, w_o^{2a}(1 + |\nabla T_o|/w_o)^2,$$

and the exponent $2 - 2a$ is positive because $a < 1$. The tips have codimension five in the link, and the weight $\tilde{w}^{2a} \sim (\sin t)^{-2a}$ is far too mild to see them. The weighted Bochner estimate on Ω_o makes the y-integral finite, the same dominated convergence argument applies, and the left side of (9.1) converges to $\varepsilon \int_0^\pi (\sin t)^{4-2a}\,\mathrm{d}t \int_{\Omega_o} w_o^{2a} > 0$.

Theorem 11.2 (Exclusion in dimension five)**.** *If a certificate on $\mathbb{R}^5$ exists, there is no nonflat minimizing cone with an isolated singularity in $\mathbb{R}^5$.*

12 Proof of the main theorem and consequences

We can now assemble the induction. Write $\mathrm{Flat}(n)$ for the statement that every nonzero minimizing cone in $\mathbb{R}^n$ is flat. By Proposition 2.3, $\mathrm{Flat}(n-1)$ together with the nonexistence of nonflat minimizing cones with an isolated singularity in $\mathbb{R}^n$ implies $\mathrm{Flat}(n)$. Hence

$$\mathrm{Flat}(1) \overset{S^1}{\Longrightarrow} \mathrm{Flat}(2) \overset{\mathbb{R}^2}{\Longrightarrow} \mathrm{Flat}(3) \overset{\mathbb{R}^3}{\Longrightarrow} \mathrm{Flat}(4) \overset{\mathbb{R}^5,\ \text{suspension}}{\Longrightarrow} \mathrm{Flat}(5) \overset{\mathbb{R}^5}{\Longrightarrow} \mathrm{Flat}(6),$$

where the label indicates the trace-free 1×1 argument or the certificate used in Theorem 9.3 (Theorem 11.2 for the step to Flat(5)). This proves the first sentence of Theorem 1.1; the second is [2].

The singular set. Corollary 1.2 follows from Weiss's dimension reduction [11] (see [12]): blowups at singular points are nonflat minimizing cones, blowups of those away from the origin split off lines, and the absence of nonflat cones below dimension seven gives $\mathrm{Sing}(u) = \varnothing$ for $n \le 6$, local finiteness for $n = 7$, and $\dim_{\mathcal{H}} \mathrm{Sing}(u) \le n - 7$ in general. For sharpness, let U be the De Silva–Jerison cone and consider $(x, y) \mapsto U(x)$ on $\mathbb{R}^7 \times \mathbb{R}^{n-7}$. By Lemma 11.1 it is a minimizer, every point of $\{0\} \times \mathbb{R}^{n-7}$ is singular since every rescaling there is the function itself, and this set has dimension $n - 7$.

13 Where the dimension enters

The proof shows that dimension seven is sharp not only for the problem but for the method, within the class of certificates of Definition 7.6 (any coefficients, any exponent $a \in (1/2, 1)$). A certificate on $\mathbb{R}^6$ would, by Theorem 9.3 with $n = 7$, exclude all nonflat minimizing cones with an isolated singularity in $\mathbb{R}^7$. Since all cones in $\mathbb{R}^6$ are flat, every minimizing cone in $\mathbb{R}^7$, in particular the cone of De Silva and Jerison, has an isolated singularity (Proposition 2.3). Hence no certificate on $\mathbb{R}^6$ exists: for every choice of coefficients, the interior or the boundary inequality fails somewhere on the finite-dimensional space of jets. This is the analogue of the obstruction $\sin(\pi x/L)$ in Example 4.1.

Where exactly the dimension enters is visible in the structure of the inequalities. The interior polynomial contains $mU_{jk}A_{jk}$, from $\Delta T = mT$, against $\beta_m^2 f = ((m-1)/2)^2 f$ from the Hardy inequality: a term linear in m against one quadratic in m. The improved Kato constant, $m/(m+2)$ in general, deteriorates with m. And the set of boundary shapes grows, admitting from $m = 4$ on configurations with κ near 0 and $\operatorname{tr} A^3 > 0$ on which plain powers fail. The examples of Section 6 show the window closing for the simplest ansatz already at $m = 3$. The certificates show that a richer ansatz reopens it at $m = 2$, 3 and 5 ($m = 4$ being bypassed by the suspension), and the De Silva–Jerison cone shows that nothing reopens it at $m = 6$.

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